

by Shaowei Chu and Jiro Tanaka at the University of Tsukuba, Japan.

The camera applies computer vision algorithm to recognise user's hand gestures. The system works with cameras equipped with viewfinders that can be flipped round to face the front, or those that have a viewfinder on the front and the back. This lets the users see the frame and control virtual buttons that appear on the screen by moving their hand over where they appear in the frame.

The pan and tilt platform guarantees that the camera focuses on the user automatically. The large display shows the augmented live-view video, which gives a clear preview to user. By using hand gestures, the user can control the camera functions from a distance.

This is just a primary attempt. Numerous interfaces can be implemented, such as the slider bar, clock move interface and button. Moreover, if the interface is set on the region of the face, the head-shake and nod gesture can be recognised as well.

A T-shirt that charges mobile phones

Here is a new T-shirt from Orange that charges mobile phones while one watches musical bands in action. All you need to do is to plug in your phone into the T-shirt for a quick top-up charge whenever it needs it.

The futuristic garment works by using noise-responsive technology; louder the music, quicker the phone charges, say its developers.

It uses an A4-size piece of piezoelectric film—a technology commonly used in hi-fi speakers—to absorb pressure from sound waves. The energy gathered from sound waves is stored in a reservoir battery in the form of electrical charge, which it then transfers from its battery into a lead that fits most phones.

New cooling pump from NASA

Thermal control is always one of the limiting factors, particularly in space



PossessedHand

where there is no air to help cool down electronic components. However, a new prototype pump from NASA is likely to clear the hurdle.

Jeffrey Didion, a thermal engineer at the NASA Goddard Space Flight Center in Greenbelt, Maryland and Dr Jamal Seyed-Yagoobi, a professor at the Illinois Institute of Technology in Chicago, have collaborated to develop a technology that may overcome current limitations.

They have formed technical partnerships with the US Air Force and National Renewable Energy Laboratory to address the thermal-control concerns.

Called electrohydrodynamic (EHD)-based thermal control, the technology promises to make it easier and more efficient to remove heat from small spaces—a particular challenge for engineers building advanced space instruments and microprocessors that could fail if the heat they generate is not removed.

The main objective of the EHD demonstration is showing that a prototype pump can withstand the extreme launch loads as the rocket lifts off and hurtles toward space. "Should it survive the vibration, the technology will have achieved a major milestone in its development," said Didion. It will mean that it is at or near operational status, making it a viable technology for use on space-

flight instruments.

World's smallest 3D printer

Researchers at Vienna University of Technology have developed what they claim to be the world's smallest 3D printer. The new printer is the size of a milk carton and weighs 1.5 kg.

The new printer will provide rapid fabrication in smaller size and lower price needed by the home consumers. The objects are printed with a special type of resin that is light-sensitive, and hardens when it is hit by a beam of intense light. Intricate designs can be built layer by layer. This process is called additive manufacturing technology.

As the desired parts of a layer harden, the remaining resin is irradiated and a new layer can be added. These layers are just one-twentieth of a millimetre thick and are printed with such precision that the machine could be used for manufacturing detailed objects such as hearing aid parts.

World's thinnest glass for displays

Recent shifts towards increasingly thin and energy saving displays and organic light-emitting diode (OLED) lighting are now being taken to the next level with the development of next-generation flexible displays and OLED lighting.